# Stroke Risk Analysis dashboard

**5,110 Patient Records | Exploratory Risk Analysis**

**Total Patients=**
***5110***

**Total stroke cases=**
***249***

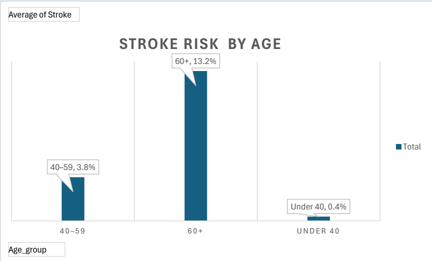
**Stroke rate=**
***4.9%***

**Average Age=**
***43.2***

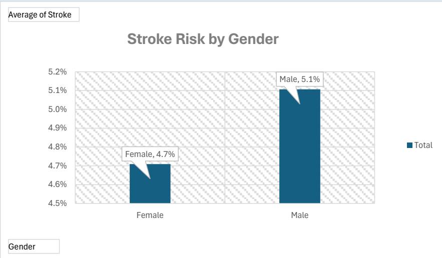
**Gender**
- Female
- Male
- Other

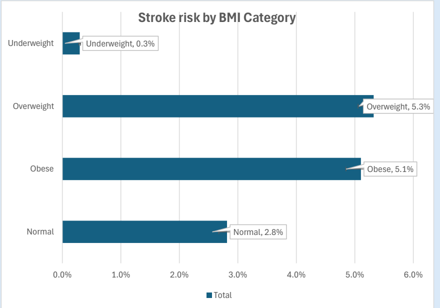
**Smoking_s...**
- formerly smoked
- never smoked
- smokes
- Unknown

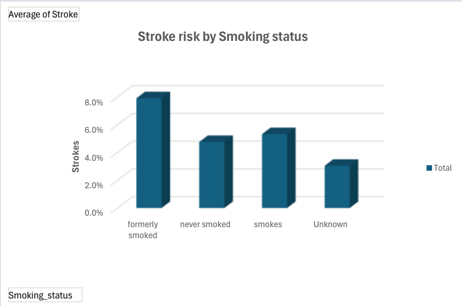
**Age_gro...**
- 40–59
- 60+
- Under 40

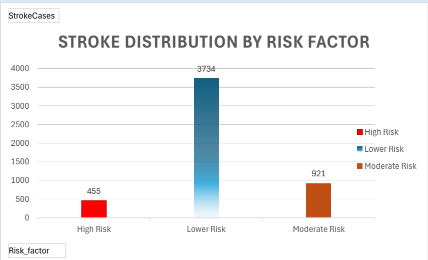



**Patients aged 60+ show over 30× higher stroke probability compared with individuals under 40.**




**Risk categories derived from combined risk factors including age, BMI, smoking status and hypertension indicators.**

***- Age is the strongest risk indicator***
*Stroke risk increases substantially with age. Patients aged* ***60+ demonstrate the highest stroke probability****, confirming age as one of the most significant predictors of stroke occurrence.*

***- Smoking history is associated with higher stroke risk***
*The analysis shows that* ***former smokers have the highest stroke probability (~8%)****, followed by current smokers (~5%). This pattern suggests the* ***long-term vascular effects of smoking may persist even after cessation****.*

***- Elevated BMI categories show higher stroke probability***
*Patients classified as* ***overweight and obese display higher stroke risk compared with individuals in the normal BMI range****, indicating body weight as a potential contributing risk factor.*

***- Overall stroke prevalence in the dataset is relatively low but meaningful***
*Out of* ***5,110 patient records, 249 stroke cases were observed****, corresponding to an overall* ***stroke rate of approximately 4.9%*** *in the population.*

***- Gender differences in stroke risk are minimal***
*Stroke probability is* ***similar between male (~5.1%) and female (~4.7%) patients****, suggesting that* ***gender alone does not appear to be a dominant determinant of stroke risk within this dataset****.*

***Conclusion:****The exploratory analysis suggests that* ***age, smoking history, and BMI are key contributors to stroke risk****, while gender appears to have limited influence in this dataset.*

***Recommendations***

• Identify high-risk patients (age 60+, smokers, high BMI) and implement early alert systems for healthcare providers.
• Develop targeted lifestyle intervention programs focusing on smoking cessation, weight management, and preventive care.

Dataset: Kaggle Stroke Prediction Dataset
Records: 5,110 patients
Analysis type: Exploratory risk factor analysis
Tool: Microsoft Excel